Submodule 39.03: Core Computer Vision Tasks aur Real-World Applications (Hinglish Guide)

1. Computer Vision Ke 4 Mukhya Tasks

Computer Vision mein alag-alag complexities ke tasks hote hain—tasveer ko sirf ek category dena se lekar har ek pixel ko classify karne tak:

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Coarse (Poori Image) ----------------------------------> Fine (Har Ek Pixel)
[ Image Classification ] -> [ Classification + Loc ] -> [ Object Detection ] -> [
```

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2. Char Mukhya Tasks Ki Paribhasha

1. Image Classification (Tasveer Pehchanna)

- **Sawaal**: *"Is image mein kaun sa object hai?"*
- **Output**: Ek single category label (jaise `'Cat'`).
- **Detail**: Isme object kahan hai, uski position predict nahi ki jati; sirf yeh bataya jata hai ki tasveer kis cheez ki hai.

2. Classification + Localization (Object + Position)

- **Sawaal**: *"Object kya hai aur tasveer mein kahan par hai?"*
- **Output**: Class label + **1 Bounding Box** coordinates (x, y, w, h):
    - (x, y): Bounding box ka top-left corner.
    - w, h: Bounding box ki chaudai aur unchai.

3. Object Detection (Multiple Objects Ko Dhoondhna)

- **Sawaal**: *"Tasveer ke andar kitne objects hain, kahan hain, aur kya hain?"*
- **Output**: Multiple bounding boxes aur unke labels (jaise ek hi photo mein Billi, Kutta aur Batakh ko alag-alag boxes mein mark karna).
- **Popular Models**: YOLO, Faster R-CNN.

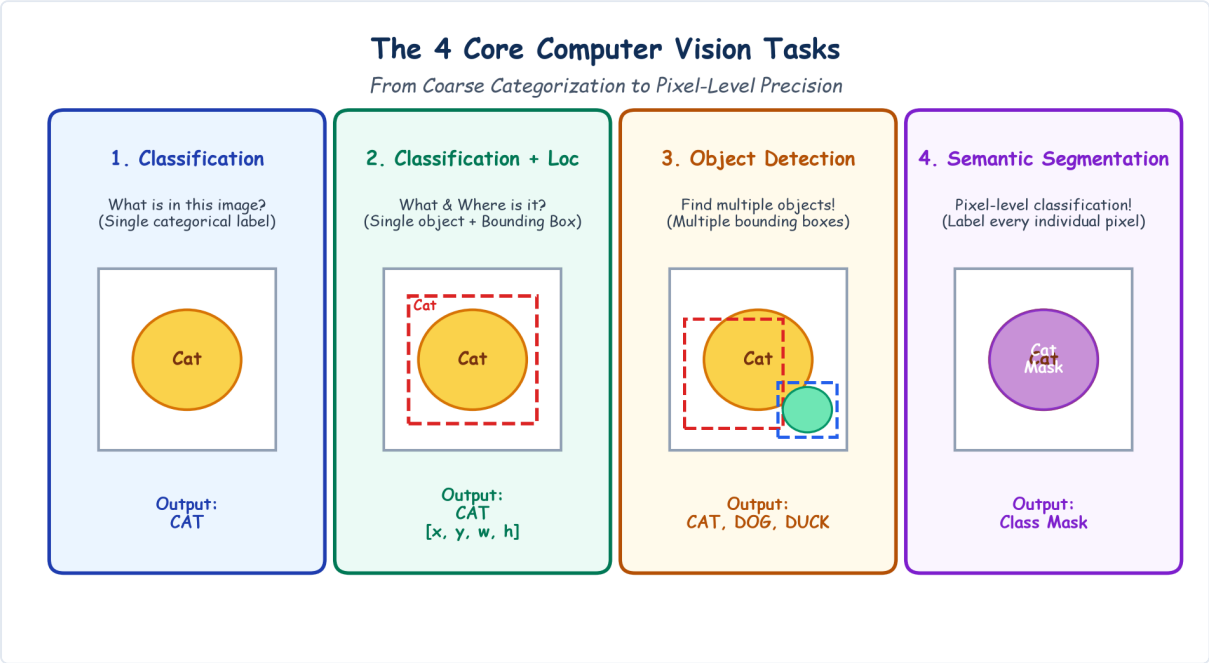
4. Semantic Segmentation (Pixel-Level Masking)

- **Sawaal**: *"Tasveer ka har ek pixel kis class se belong karta hai?"*
- **Output**: Ek 2D mask jahan har single pixel ko label milta hai (jaise Sadak ke pixels alag rang ke, Car ke alag, Pedestrian ke alag).
- **Popular Models**: U-Net, Mask R-CNN.

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3. Hand-Drawn Educational Diagram: Computer Vision Tasks

Neeche diya gaya student-style handwritten diagram charon tasks ke visual differences ko clear boxes aur labels ke sath darshata hai:



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4. Real-World Mein Computer Vision Ke Practical Examples

1. **Smartphone Face Unlock**:
    - Camera chehre ke landmarks ko scan karke phone ko unlock karta hai.
2. **Automated Traffic Challan (ALPR)**:
    - High-speed cameras gaadiyon ki number plate detect karke automatic challan generate karte hain.
3. **Self-Driving Cars (Autonomous Vehicles)**:
    - Tesla aur Waymo live video stream se road, pedestrians, stop signs aur potholes ko detect karte hain.
4. **Smart Retail (Amazon Go)**:

- Bina cashier ke store jahan cameras detect karte hain ki customer ne shelf se kaun sa product uthaya.

5. **Medical Diagnosis**:
    - X-rays, MRI scans aur CT scans mein pneumonia, fractures ya tumors ko detect karna.
6. **Instagram & Snapchat Filters**:
    - Chehre ke movements ke sath align hone wale AR 3D face filters.